

FAST National University

School of Computing — Karachi Campus

Object Oriented Programming — CS1004



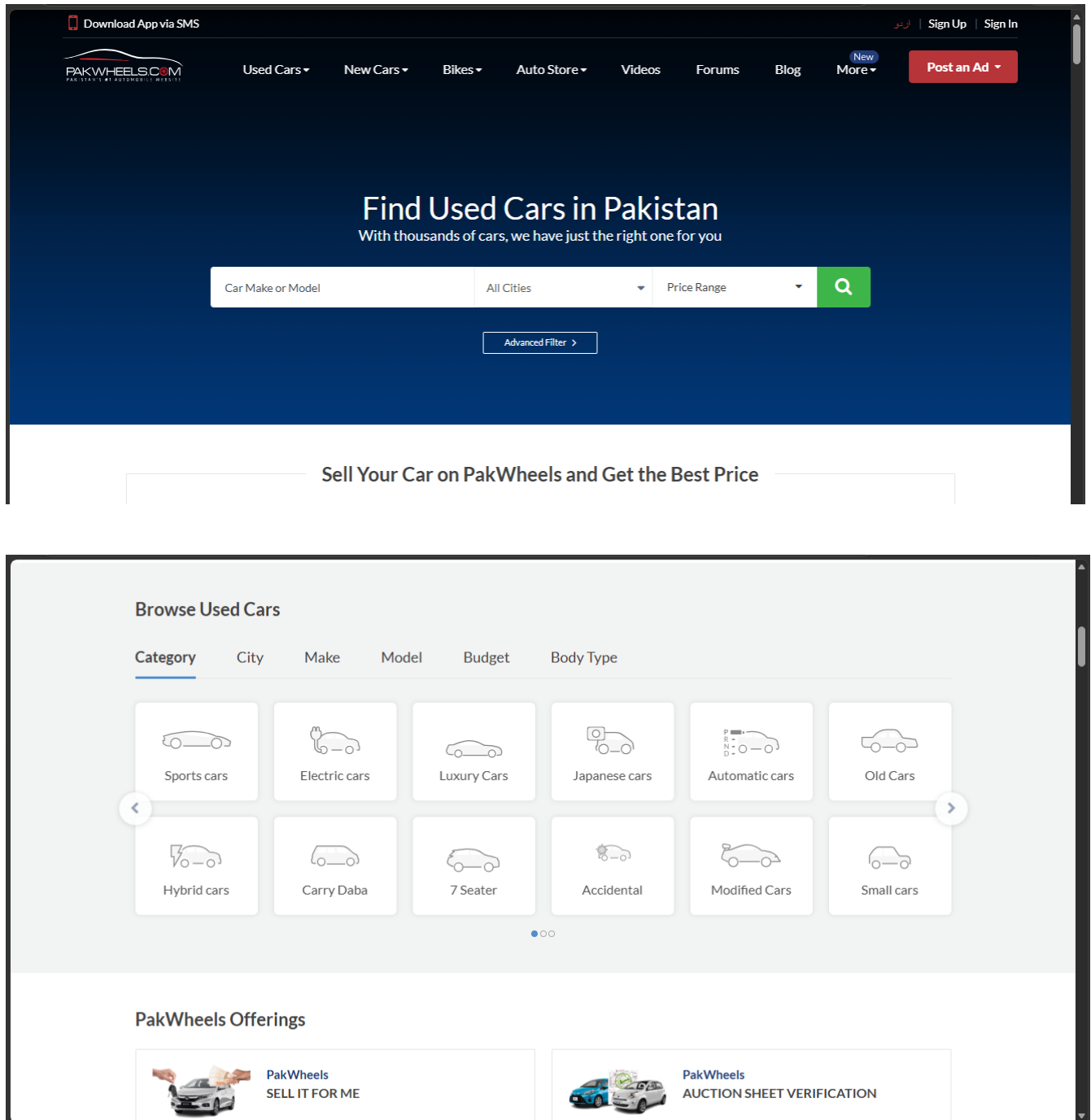
Assignment #2 Documentation

PakWheels Car Marketplace System

Semester	Spring 2026
Submitted To	Mr. Talha Shahid
Submitted On	15-05-2026
Language	C++
Reference Website	www.pakwheels.com

1. Introduction & Website Overview

PakWheels (www.pakwheels.com) is Pakistan's largest online car marketplace, connecting buyers, sellers, and dealerships. This project replicates its core domain model using OOP in C++. The classes below map directly to real website features.

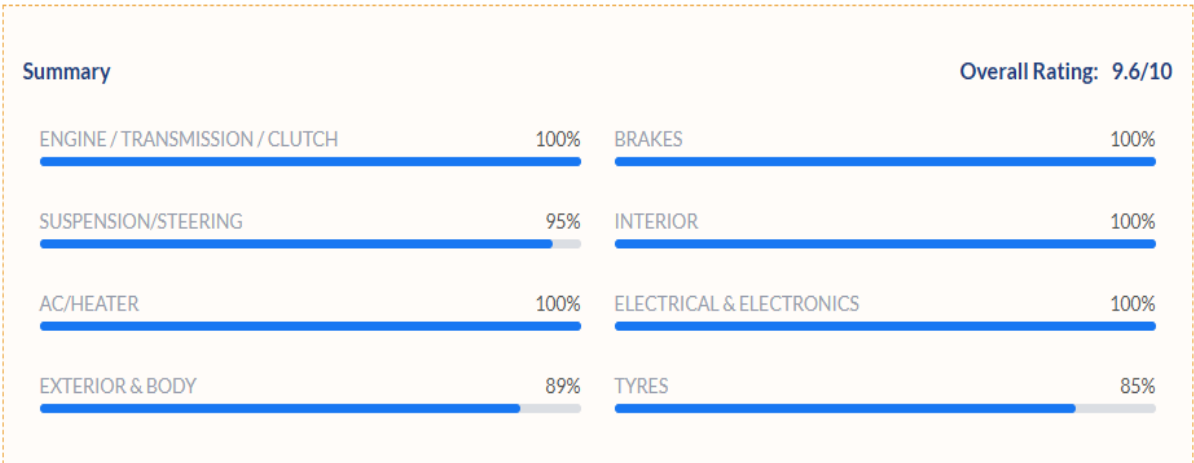


PakWheels Feature	Classes	OOP Concept
Car Listings	Vehicle, Car, Bike, ElectricCar	Inheritance, Array of Objects
Inspection Badge	InspectionReport, StandardReport, Vehicle	Abstraction, Composition
User Accounts	User, Buyer, Seller, Admin	Inheritance, Polymorphism
Saved / Shortlisted Cars	Favorite, Buyer	Composition, Array of Objects
Buyer-Seller Messaging	Message, Buyer	Composition, Array of Objects
Seller Ratings & Reviews	Review, Seller	Aggregation, Array of Objects
Admin Listing Approval	Admin, PakWheelsSystem	Polymorphism, Friend Functions
Search & Filter	PakWheelsSystem, Vehicle	Static Members, Const Functions
EV Listings	ElectricCar	Multi-level Inheritance

2. Class-by-Class Documentation

2.1 InspectionReport (Abstract) & StandardReport — Composition inside Vehicle

PakWheels offers a paid CarSure inspection service; inspected cars receive a verification badge on the listing. In this version, InspectionReport is an abstract base class with pure virtual print() and generateSummary(). StandardReport is the concrete subclass that stores the four inspection category scores.



InspectionReport — Abstract Base Class

Data Members	Member Functions
bool passed	InspectionReport() — default
string notes	InspectionReport(bool p, string n) — param
static int totalInspections	InspectionReport(other) — copy
	virtual void print() const = 0
	virtual string generateSummary() const = 0
	void set(bool p, string n)
	bool hasPassed() const
	static int getTotal()
	bool operator==(const InspectionReport&)

StandardReport — Concrete Subclass

Data Members	Member Functions
int exterior	StandardReport() — default
int interior	StandardReport(e,i,en,el,n) — param
int engine	StandardReport(other) — copy
int electrical	void set(int e, int i, int en, int el, string n)
	void print() const override
	string generateSummary() const override
	int getExterior/Interior/Engine/Electrical() const

- InspectionReport is abstract — print() and generateSummary() are pure virtual, forcing every report type to define its own output format.
- StandardReport is composed inside Vehicle — a report is meaningless without its car; destroyed with the vehicle.
- Pass condition: all four scores ≥ 6 . Any score below 6 sets passed = false.
- static totalInspections is a platform-wide analytics counter shared across all instances.
- using InspectionReport::set exposes the base set(bool, string) alongside the overloaded set(int,int,int,int,string) in the same scope.

```
C InspectionReport.h > ...
1  #pragma once
2  #include <iostream>
3  #include <string>
4  using namespace std;
5
6  class InspectionReport {
7      protected:
8          bool passed;
9          string notes;
10         static int totalInspections;
11     public:
12         InspectionReport();
13         InspectionReport(bool p, string n);
14         InspectionReport(const InspectionReport& o);
15         virtual ~InspectionReport();
16
17         virtual void print() const = 0;
18         virtual string generateSummary() const = 0;
19
20         void set(bool p, string n);
21         bool hasPassed() const;
22         static int getTotal();
23         bool operator==(const InspectionReport& o) const;
24     };
25
26     class StandardReport : public InspectionReport {
27     private:
28         int exterior;
29         int interior;
30         int engine;
31         int electrical;
32     public:
33         StandardReport();
34         StandardReport(int e, int i, int en, int el, string n);
35         StandardReport(const StandardReport& o);
36
37         using InspectionReport::set;
38         void set(int e, int i, int en, int el, string n);
39         void print() const override;
40         string generateSummary() const override;
41         int getExterior() const;
42         int getInterior() const;
43         int getEngine() const;
44         int getElectrical() const;
45     };
```

2.2 Vehicle — Abstract Listing Base Class

Every car listed on PakWheels is a Vehicle. The listing page shows brand, model, year, price, mileage, and a Verified badge if inspected. Vehicle is abstract — `displayDetails()` is pure virtual, implemented differently by Car, Bike, and ElectricCar.

Honda Civic RS 2025

📍 Canal Town, Peshawar KPK

FEATURED









2025



13,323 km



Petrol



Automatic

Registered In	Islamabad	Color	Crystal Black Pearl
Assembly	Local	Engine Capacity	1500 cc
Body Type	Sedan	Last Updated:	May 15, 2026
Ad Ref #	11345029		

PKR 98.5 lacs

Financing starts at PKR 101,916/Month

Managed by PakWheels

Buy Managed By PakWheels Car with Trust



Inspected By PakWheels



Documents Checked



Secure Transaction

042XXXXXXX
Show Phone Number

Send Message

Seller Details

TRUSTED SELLER

PAKWHEELS.COM

Dealer: PakWheels Islamabad ✓

Address: Office # 20, 2nd Floor, I-8 Markaz, Islamabad (Near Meezan Bank)

Timings: 09:00 AM to 09:00 PM

[more ads by PakWheels Islamabad »](#)



See if your friends know this seller
[Connect with facebook](#)

Data Members	Member Functions
int vId, year, mileage	Vehicle() — default
string brand, model	Vehicle(id,b,m,y,p,km) — param
double price	Vehicle(other) — copy
bool approved, deleted	virtual void displayDetails() const = 0
StandardReport report (composed)	void approve() / remove()
static int totalListings	void setPrice() / addReport()
	bool matchBrand/Model/underPrice/etc. const
	static int getTotal()
	operator==, operator+, operator<

```

bool Vehicle::operator==(const Vehicle& o) const {
    if (vId == o.vId) {
        return true;
    }
    return false;
}

double Vehicle::operator+(const Vehicle& o) const {
    return price + o.price;
}

bool Vehicle::operator<(const Vehicle& o) const {
    if (price < o.price) {
        return true;
    }
    return false;
}

```

- displayDetails() is pure virtual — Car, Bike, and ElectricCar each render their listing differently.
- remove() soft-deletes (deleted=true) rather than erasing, preserving data integrity.
- operator+ returns combined price (double). operator< compares price. operator== compares vId.
- All filter functions are const — they only read state, never modify it.
- Copy constructor required because PakWheelsSystem stores Vehicle pointers via new Car(c) etc.

2.3 Car, Bike, ElectricCar — Concrete Vehicle Subclasses

PakWheels supports regular petrol/diesel cars, motorbikes, and an EV section (ElectricCar extends Car for multi-level inheritance). Each overrides displayDetails() to print its type-specific fields.

BYD Atto 3 Advance 2025

Islamabad Islamabad | Added via Phone

FEATURED





2025

19,000 km

Electric

Automatic

Registered In	Islamabad	Color	Ski White
Assembly	Imported	Battery Capacity	49.0 kWh
Body Type	Crossover	Last Updated:	May 14, 2026

PKR 89.75 lacs

032XXXXXXXX

Show Phone Number

Send Message

Seller Details

Uzair Sheraz
Member Since Mar 12, 2020



See if your friends know this seller
Connect with facebook

Safety tips for transaction

1. Use a safe location to meet seller

2. Avoid cash transactions

3. Beware of unrealistic offers

Learn More

Notify As Sold

Report This Ad

Class	Extra Data Members	Key Functions
Car	int doors, string fuelType	Car(id,b,m,y,p,km,d,ft) getDoors(), getFuelType() displayDetails() override
Bike	int cc	Bike(id,b,m,y,p,km,cc) getCC() displayDetails() override
ElectricCar	int rangeKm, double batteryKwh	ElectricCar(id,b,m,y,p,km,range,bat) getRangeKm(), getBatteryKwh() displayDetails() override

- ElectricCar extends Car (multi-level: ElectricCar -> Car -> Vehicle). fuelType is hardcoded to "Electric" in the constructor.
- displayDetails() skips output if deleted == true — enforces soft-delete visibility.
- Inspection report line is only printed if approved == true — mirrors PakWheels' verified badge logic.


```

class Car : public Vehicle {
protected:
    int doors;
    string fuelType;
public:
    Car();
    Car(int id, string b, string m, int y, double p, int km, int d, string ft);
    Car(const Car& o);
    void displayDetails() const override;
    int getDoors() const;
    string getFuelType() const;
};

class Bike : public Vehicle {
private:
    int cc;
public:
    Bike();
    Bike(int id, string b, string m, int y, double p, int km, int engineCC);
    void displayDetails() const override;
    int getCC() const;
};

class ElectricCar : public Car {
private:
    int rangeKm;
    double batteryKwh;
public:
    ElectricCar();
    ElectricCar(int id, string b, string m, int y, double p, int km, int range, double battery);
    void displayDetails() const override;
    int getRangeKm() const;
    double getBatteryKwh() const;
};

```

2.4 Review — Aggregated by Seller

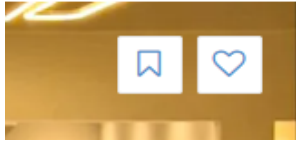
Buyers leave star ratings and comments on seller profiles. Review stores rating (1–5), comment, buyer name, and date.

Data Members	Member Functions
int rating	Review() — default
string comment	Review(r,c,bn,d) — param
string buyerName	Review(other) — copy
string date	int getRating() const
bool verified	bool isVerified() const
	bool isPositive() const
	void show() const

- Aggregated by Seller — reviews are written by Buyers externally and could be archived even after a seller account closes.
- verified mirrors PakWheels' verified-buyer badge on reviews.
- isPositive() returns true if rating ≥ 4 .

2.5 Favorite — Composed inside Buyer

PakWheels has a Shortlist (heart icon) feature where buyers save cars. Favorite stores the vehicle ID, brand/model, a personal note, and the price at time of saving.



Data Members	Member Functions
int vehicleId	Favorite() — default
string brand	Favorite(id,b,m,n,p) — param
string model	int getId() const
string note	double getSavedPrice() const
double savedPrice	bool priceRisen(double current) const
	void updateNote(string n)
	void show() const

- Composed inside Buyer — saved cars are personal and deleted with the buyer account.
- savedPrice records the price at save time, enabling a 5% price-rise alert via priceRisen().

2.6 Message — Composed in Buyer Inbox

PakWheels provides built-in chat between buyers and sellers. Messages store sender ID, receiver ID, content, timestamp, and read status.



Call for price



Data Members	Member Functions
int msgId	Message() — default
int fromId, toId	Message(id,c,f,t,ts) — param
string content	Message(other) — copy
string timestamp	void markRead()
bool read	int getToId() / getFromId() const
	bool isUnread() const

	bool involves(int id) const
	void show() const

- Copy constructor required — messages are copied into the Buyer inbox array on delivery.
- involves(id) filters conversation history for a given user ID.

2.7 User — Abstract Base Class

All PakWheels accounts share common attributes. User is the abstract base class with pure virtual performRole() forcing all subclasses to define their behavior.

Data Members	Member Functions
int userId	User() — default
string name, email, phone	User(id,n,e,ph,t) — param
char type ('B','S','A')	virtual ~User()
bool active	virtual void performRole() const = 0
const string country = "Pakistan"	virtual void showProfile() const
static int totalUsers	void deactivate()
	int getId(), getName(), getType() const
	static int getTotalUsers()
	bool operator==(const User&) const

- performRole() is pure virtual — enforces polymorphism across all account types.
- const string country = "Pakistan" — PakWheels is Pakistan-only; every user is local.
- static totalUsers is shared across Buyer, Seller, Admin; incremented in constructor, decremented in destructor.

2.8 Buyer — extends User

Buyers browse listings, shortlist cars, and message sellers. Buyer owns two composed arrays: a Favorites array (up to 10) and an inbox (up to 20 messages).

Data Members	Member Functions
Favorite favs[10] (COMPOSITION)	Buyer(id,n,e,ph,b) — param
Message inbox[20] (COMPOSITION)	void addFavorite(vid,b,m,note,p)
int favCount, msgCount	void removeFavorite(int vid)
double budget	void receiveMsg(const Message& m)
const int maxFavs = 10	void viewInbox() const
	void viewFavorites() const
	void performRole() const override
	void showProfile() const override
	double getBudget() / setBudget() const

- favs[10] is composition — favourites are created and destroyed with the Buyer.
- const int maxFavs enforces the platform's per-account saved-car limit.
- removeFavorite() manually shifts the array (no STL) to fill the gap.
- PakWheelsSystem is a friend class — allows sendMessage() to call receiveMsg() directly.

2.9 Seller — extends User

Sellers post car ads and accumulate buyer reviews. The reviews array is aggregation because reviews are written by buyers and added externally.

Data Members	Member Functions
int listings[10] (vehicle IDs)	Seller(id,n,e,ph) — param
Review reviews[20] (AGGREGATION)	void listCar(int vid)
int listCount, reviewCount	bool delistCar(int vid)
double revenue	void addReview(const Review& r)
const int maxListings = 10	void addRevenue(double a)
	double avgRating() const
	int getListCount() const
	void showReviews() const
	void performRole() const override

- Review[] is aggregation — reviews are passed in from Buyers and represent external feedback.
- const int maxListings = 10 mirrors PakWheels' free listing limit per seller.
- avgRating() computes dynamically to avoid storing redundant state.
- auditRevenue() is a friend function — accesses private revenue and listCount for admin auditing.

2.10 Admin — extends User

The PakWheels admin team approves inspected cars and removes fraudulent ads. Admin has elevated privileges, with a second static counter (totalAdmins).

Data Members	Member Functions
string level (e.g. "SuperAdmin")	Admin(id,n,e,ph,lv) — param
int approved, removed	~Admin() — decrements totalAdmins
bool canBan	void approveCar(Vehicle& v)
string department	void approveCar(Vehicle& v, string note)
static int totalAdmins	void removeCar(Vehicle& v)
	void banUser(User& u)
	void performRole() const override
	int getApproved() / getRemoved() const
	static int getTotalAdmins()

- static totalAdmins is a second static member — separate from totalUsers, counts only active admins.
- canBan is set by level — only SuperAdmin can ban users, mirroring role-based access control.
- ~Admin() decrements totalAdmins, keeping the count accurate on object deletion.
- approveCar() is overloaded — one version without a note, one with a note string (function overloading).

2.11 PakWheelsSystem — Central Controller

The central platform class owns the vehicle inventory, holds polymorphic user pointers, logs messages, and provides all search and filter operations.

Data Members	Member Functions
Vehicle* inventory[50]	PakWheelsSystem() / ~PakWheelsSystem()
User* users[30] (AGGREGATION)	void registerUser(User* u)
Message msgLog[100]	void addCar/Bike/EV(...)
int vCount, uCount, msgCount	Vehicle* findVehicle(int id)
const string name = "PakWheels"	User* findUser(int id)
static int totalSearches	bool updatePrice(int vid, double p)
	searchBrand / searchModel const
	filterPrice / filterMileage / filterYear const
	void sendMessage(...)
	void showStats() const
	static int getSearches()
	friend compareVehicles / listAllVehicles

- User* users[30] is aggregation — users are created outside and registered; not owned by the system.
- Vehicle inventory stores pointers (new Car/Bike/ElectricCar) — destructor deletes them, preventing memory leaks.
- static totalSearches — platform analytics counter, incremented on every search/filter call.
- const string name = "PakWheels" — the platform name is fixed and never changes.
- compareVehicles() and listAllVehicles() are friend functions — they access private inventory and vCount directly.
- sendMessage() uses a C-style cast to Buyer* to deliver to the buyer inbox — runtime polymorphism via getType().

3. OOP Concepts — Evidence & Reasoning

3.1 Constructors

Constructor Type	Used In	Reason
Default	InspectionReport, StandardReport, Vehicle, Car, Bike, ElectricCar, Review, Favorite, Message, User	Required when objects are stored in arrays — all array elements are default-constructed at declaration.
Parameterized	All classes	The main constructor for creating meaningful objects with real data.
Copy	InspectionReport, StandardReport, Vehicle, Car, ElectricCar, Message	Vehicle is stored via new Car(c) in PakWheelsSystem; InspectionReport is a composed member auto-copied. Messages are copied into the Buyer inbox array.

3.2 Constants & Constant Functions

#	Constant / Const Function	Reason
1	const string User::country = "Pakistan"	PakWheels is Pakistan-only; user country is fixed at construction.
2	const int Buyer::maxFavs = 10	Platform cap on saved cars per buyer account.
3	const int Seller::maxListings = 10	Free listing limit per seller account.
4	const string PakWheelsSystem::name = "PakWheels"	The platform name is fixed and never changes.
5	bool InspectionReport::hasPassed() const	Read-only — does not modify any member.
6	void Vehicle::displayDetails() const	Displays info without modifying vehicle state.
7	double Seller::avgRating() const	Computes average from reviews without modifying data.
8	Vehicle filter functions (underPrice, fromYear, etc.) const	Search and filter functions only read state — never modify it.

3.3 Static Members

#	Static Member	Reason
1	User::totalUsers	Counts all registered accounts; shared across Buyer, Seller, Admin. Incremented in constructor, decremented in destructor.
2	InspectionReport::totalInspections	Counts platform-wide inspections — a business analytics metric.
3	Vehicle::totalListings	Total vehicle objects ever created — platform listing counter.
4	Admin::totalAdmins	Separate count of active admin accounts, independent of totalUsers.
5	PakWheelsSystem::totalSearches	Incremented on every search/filter call — web analytics.

3.4 Composition & Aggregation

Type	Relationship	Reasoning
------	--------------	-----------

Composition	Vehicle owns StandardReport report	A report is meaningless without its vehicle. Deleted together.
Composition	Buyer owns Favorite favs[10]	Saved cars belong exclusively to one buyer; destroyed with account.
Composition	Buyer owns Message inbox[20]	A buyer's inbox is private and lives with the buyer account.
Aggregation	Seller aggregates Review reviews[20]	Reviews are created by Buyers externally — could be archived independently.
Aggregation	PakWheelsSystem aggregates User* users[30]	Users are created outside and registered; system does not own them.

3.5 Array of Objects

Array	Class	Purpose
Vehicle* inventory[50]	PakWheelsSystem	Stores polymorphic vehicle pointers; iterated for search and filter.
Favorite favs[10]	Buyer	Each buyer's saved cars — composed array.
Message inbox[20]	Buyer	Buyer's received messages — composed array.
Review reviews[20]	Seller	Seller's buyer reviews — aggregated array.
Message msgLog[100]	PakWheelsSystem	Platform-wide message log for all sent messages.

3.6 Polymorphism & Inheritance

User is the abstract base with pure virtual performRole(). Buyer, Seller, and Admin each override it. PakWheelsSystem stores User* pointers and dispatches performRole() polymorphically at runtime. Similarly, Vehicle is abstract with pure virtual displayDetails() — Car, Bike, and ElectricCar provide their own implementations.

Subclass	performRole() Override
Buyer	Prints name + budget, simulating a buyer browsing the site.
Seller	Prints name + active listing count.
Admin	Prints name, level, approved and removed counts.

Subclass	displayDetails() Override
Car	Prints [Car] tag, brand, model, year, doors, fuel type, price, mileage, approval status, and report if verified.
Bike	Prints [Bike] tag with engine cc instead of doors/fuel.
ElectricCar	Prints [EV] tag with rangeKm and batteryKwh fields.

3.7 Operator Overloading

Operator	Class	Behavior
operator==	Vehicle	Returns true if vld matches — same listing check.
operator<	Vehicle	Returns true if price is lower — cheaper vehicle.

operator+	Vehicle	Returns combined price (double) of two vehicles.
operator==	User	Returns true if userId matches.
operator==	InspectionReport	Returns true if passed status matches.

3.8 Friend Functions

Friend	Granted By	Purpose
class PakWheelsSystem	Buyer	Allows sendMessage() to call receiveMsg() on a Buyer directly.
void auditRevenue(const Seller&)	Seller	Accesses private revenue and listCount for admin auditing.
void compareVehicles(...)	PakWheelsSystem	Accesses private inventory[] and vCount for side-by-side comparison.
void listAllVehicles(...)	PakWheelsSystem	Accesses private inventory[] and vCount for full admin inventory dump.

```
void auditRevenue(const Seller& s) {
    cout << "  Audit -> " << s.name << " | Revenue: PKR " << s.revenue << " | Listings: " << s.listCount << "\n";
}
```

4. Program Output

Below is the expected console output when main.cpp is compiled and run with the provided source files.

```
User Roles
Ahmed Khan managing 0 listing(s).
Bilal Raza managing 0 listing(s).
Sara Malik browsing PakWheels under PKR 5.5e+06
Usman Tariq browsing PakWheels under PKR 1.2e+07
Admin Superuser [SuperAdmin] - Approved:0 Removed:0

Inventory Listing

Full Inventory (Admin View)
[Car] Toyota Corolla (2020) 4-door Petrol | PKR 4.8e+06 | 45000km [Pending]
[Car] Honda Civic (2021) 4-door Petrol | PKR 6.2e+06 | 22000km [Pending]
[Car] Suzuki Alto (2019) 4-door Petrol | PKR 2.1e+06 | 80000km [Pending]
[Car] Toyota Fortuner (2022) 4-door Diesel | PKR 9.5e+06 | 15000km [Pending]
[Car] KIA Sportage (2023) 4-door Petrol | PKR 8.75e+06 | 5000km [Pending]
[EV] Tesla Model 3 (2023) Range:500km | Battery:75.5kWh | PKR 1.5e+07 | 12000km [Pending]
[EV] BYD Atto 3 (2024) Range:420km | Battery:60kWh | PKR 1.15e+07 | 3000km [Pending]

Operator Testing
Toyota is cheaper than Honda
Combined price (101 + 102): 1.1e+07
Suzuki Alto is cheaper than Toyota Fortuner
Combined price (103 + 104): 1.16e+07

Abstraction Summaries
ID 101: PASS - Clean engine, minor scratches
ID 102: PASS - Well maintained, single owner
ID 103: FAIL - Engine okay, body rust present
ID 104: PASS - Excellent condition, low mileage
ID 105: PASS - Brand new condition
ID 301: PASS - Battery perfect, no degradation
ID 302: PASS - Minor software update needed

Affordability Check (Sara: 5,500,000)
Sara can afford: Toyota Corolla
Sara can afford: Suzuki Alto
Sara cannot afford: Honda Civic

Affordability Check (Usman: 12,000,000)
Usman cannot afford: Tesla Model 3
Usman can afford: BYD Atto 3
```

5. UML Class Diagram

The diagram below shows all classes, their inheritance hierarchy, composition, and aggregation relationships.

Inheritance Hierarchy

Base Class	Derived Classes	Relationship
User (abstract)	Buyer, Seller, Admin	Public inheritance — all override performRole()
Vehicle (abstract)	Car, Bike, ElectricCar	Public inheritance — all override displayDetails()
Car	ElectricCar	Multi-level inheritance
InspectionReport (abstract)	StandardReport	Public inheritance — overrides print() and generateSummary()

